

openheart Genetic testing in cardiomyopathies: do we need to redefine the UK national testing criteria?

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ABSTRACT

Introduction Inherited cardiac conditions, including dilated cardiomyopathy (DCM) and hypertrophic cardiomyopathy (HCM), may have a monogenic cause identified through genetic testing (GT). Confirmation of pathogenic gene variants can have important implications for the patient and their relatives. The UK National Genomic Test Directory (NGTD) provides strict criteria on the indications for GT; however, the European Society of Cardiology (ESC) recommends wider GT. We reviewed the prevalence of pathogenic genotypes in patients undergoing GT who did not meet the NGTD criteria.

Methods We conducted a retrospective analysis of patients who underwent GT with a confirmed diagnosis of HCM or DCM attending the Essex Inherited Cardiac Conditions Clinic between January 2023 and January 2025.

Results 257 patients were included in the analysis, with 136 patients with DCM (52.9%) and 121 patients with HCM (47.1%). The diagnostic yield of GT was 19.9% in DCM and 17.4% in HCM.

14.8% of gene-positive patients with DCM and 14.3% of gene-positive patients with HCM did not meet current UK testing criteria, predominantly due to age of onset. All gene-positive patients in the DCM subgroup not meeting current NGTD criteria for testing had evidence of myocardial fibrosis.

Conclusion A significant minority of patients (1 in 7) with cardiomyopathy and a pathogenic genotype did not meet current UK testing criteria; each patient has an average of 4 first-degree relatives at risk who will benefit from predictive GT. We propose the adoption of the wider ESC guidance, removing the strict age-related cut-offs and being guided more by the severity of the phenotype, particularly involving myocardial scarring.

INTRODUCTION

Among the spectrum of Inherited Cardiac Conditions (ICC), the myocardial disorders dilated cardiomyopathy (DCM) and hypertrophic cardiomyopathy (HCM) represent the most prevalent subtypes encountered in clinical practice. Although the majority of cardiomyopathy phenotypes are not due to monogenic causes, identification of a pathogenic or likely pathogenic gene variant has

important implications for first-degree relatives and may guide therapies in the proband (index case) as well as cascade screening with predictive genetic testing (GT).^{1,2}

UK National Genomic Test Directory (NGTD)

The UK NGTD established specific criteria to guide the referral and triage of patients for genomic testing across a wide range of inherited diseases.³ The NGTD is commissioned by the National Health Service (NHS) in England, and its focus is to guide requests that will have high yield for results that are highly likely to change clinical management of the patient or their family. While cost implications are not explicitly mentioned in the guidance, there is clearly a need to rationalise resources within a public healthcare system. However, strict adherence to such guidelines may exclude patients who have pathogenic variants but fall outside the defined criteria. The DCM/ACM (arrhythmogenic cardiomyopathy) R132 subpanel and HCM R131 subpanel genes and testing criteria are listed in figure 1.

Why may patients be tested outside of the criteria for DCM?

Prior to the first European Society of Cardiology (ESC) cardiomyopathy guidelines published in 2023, there was an ill-defined cardiomyopathy phenotype, now termed non-dilated left ventricular cardiomyopathy (NDLVC), which is consistent with a non-ischaemic cardiomyopathy complicated by scarring but without objective left ventricular (LV) dilatation.⁴ The guidelines clearly described the high prevalence of scar burden in NDLVC and its correlation with an underlying genetic substrate.

The absence of a family history does not exclude the possibility of a familial condition, with phenotypic heterogeneity common in some variants. There is also a possibility of familial deaths being deemed non-cardiac or



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A: Dilated and Arrhythmogenic Cardiomyopathy (DCM/ACM)

Panel Genes (R132)	Relevant NGTD Testing Criteria
TTN, LMNA, FLNC, PLN, DES, DSP, RBM20, SCN5A, ACTC1, ACTN2, BAG3, CDH2, DMD, DOLK, DSC2, DSG2, EMD, JUP, LAMP, MYBPC3, MYH7, NEXN, NKX2-5, PKP2, TMEM43, TNNC1, TNNI3, TNNI3K, TNNT2, TPM1, VCL	1. Left ventricular end diastolic diameter (LVEDD) greater than 2 standard deviations, AND/OR a. Reduced ejection fraction (EF) to less than 45%, adjusted for age and sex, AND b. Age of onset below 65 years, OR c. DCM with conduction defects, with age of onset below 65 years 2. Left and/or biventricular cardiomyopathy associated with variable degrees of myocardial dysfunction and/or myocardial fibrosis PLUS ventricular arrhythmias (including prior cardiac arrest), following exclusion of other aetiologies including inflammatory disorders 3. A deceased individual with pathologically confirmed DCM or ACM and age of onset below 65 years, suitable for post-mortem DNA analysis 4. Patient with DCM or ACM at any age, if they have a first-degree relative with a confirmed diagnosis of DCM or ACM

B: Hypertrophic Cardiomyopathy (HCM)

Panel Genes (R131)	Relevant NGTD Testing Criteria
MYBPC3, MYH7, TNNT2, TNNI3, ACTC1, ACTN2, CSRP3, FHL1, FHOD3, JPH2, LAMP2, MYL2, MYL3, PLN, PRKAG2, TNNC1, TPM1, CACNA1C (p.Arg518), MT-TI (m.4300A>G), TTR, GLA	1. An adult with wall thickness ≥ 15 mm in one or more left ventricular (LV) myocardial segments, that is not explained solely by loading conditions (principally hypertension), with age of onset below 60 2. A child under the age of 18 with LV wall thickness more than two standard deviations greater than the predicted mean (z-score > 2), where a z-score is defined as the number of standard deviations from the population mean 3. Otherwise, unexplained increased LV wall thickness ≥ 13 mm in one or more LV myocardial segments, in a patient with a first-degree relative with unequivocal disease (LVH ≥ 15 mm), where a family member with unequivocal disease is unavailable for testing 4. A deceased individual with pathologically confirmed HCM suitable for post-mortem DNA analysis

Figure 1 Gene panels and testing criteria for cardiomyopathies according to the UK NGTD. (A) Gene panel (R132)¹¹ and the testing criteria in dilated and arrhythmogenic cardiomyopathy.⁹ (B) Gene panel (R131)¹² and the testing criteria in hypertrophic cardiomyopathy.¹⁰ Titin (*TTN*), lamin A/C (*LMNA*), filamin C (*FLNC*), phospholamban (*PLN*), desmin (*DES*), desmoplakin (*DSP*), RNA-binding protein 20 (*RBM20*), sodium channel type V (*SCN5A*), actin alpha cardiac muscle 1 (*ACTC1*), actinin alpha 2 (*ACTN2*), *Bcl2*-associated athanogene 3 (*BAG3*), cadherin 2 (*CDH2*), dystrophin (*DMD*), dolichol kinase (*DOLK*), desmocollin 2 (*DSC2*), desmoglein 2 (*DSG2*), emerin (*EMD*), junction plakoglobin (*JUP*), lysosomal associated membrane protein 2 (*LAMP2*), myosin-binding protein C3 (*MYBPC3*), myosin heavy chain 7 (*MYH7*), nexilin F-actin binding protein (*NEXN*), NK2 homeobox 5 (*NKX2-5*), plakophilin 2 (*PKP2*), transmembrane protein 43 (*TMEM43*), troponin C1 slow skeletal and cardiac type (*TNNC1*), troponin I3 cardiac type (*TNNI3*), troponin I interacting kinase (*TNNI3K*), troponin T2 cardiac type (*TNNT2*), tropomyosin 1 (alpha) (*TPM1*), vinculin (*VCL*), domains 1 (*FHL1*), formin homology 2 domain containing 3 (*FHOD3*), junctophilin 2 (*JPH2*), myosin light chain 2 (*MYL2*), myosin light chain 3 (*MYL3*), protein kinase AMP-activated non-catalytic subunit gamma 2 (*PRKAG2*), calcium voltage-gated channel subunit alpha1 C, p.Arg518 variant (*CACNA1C*, p.Arg518), mitochondrially encoded tRNA isoleucine m.4300A>G variant (*MT-TI*, m.4300A>G), transthyretin (*TTR*), galactosidase alpha (*GLA*). AMP, adenosine monophosphate, NGTD, National Genomic Test Directory.

not sudden cardiac death related in error, due to incomplete investigation or in the absence of postmortem examination of deceased relatives. The clinical suspicion of an unproven but suspected familial condition sometimes prompts genotyping outside of the defined criteria.

Why may patients be tested outside of the criteria for HCM?
Patients may have a clear HCM phenotype but not meet strict criteria based on absolute wall thickness. There is increasing evidence that an absolute cut-off of 15 mm wall thickness is flawed, as it fails to account for body size indexing, or the natural apical tapering of the myocardium; thus, studies have proposed an adjustment of this

criterion.⁵ Patients presenting with an early phenotype detected surreptitiously, for example, through an ECG performed routinely for non-cardiology reasons such as preoperative screening, may be declined genotyping due to wall thickness, despite very high clinical suspicion. Missed early diagnoses may have important implications for the wider family, with up to 60% penetrance in some sarcomeric gene variants.⁶

This study describes GT in a cohort of patients with DCM and HCM in the ICC service at the tertiary Essex Cardiothoracic Centre. The NGTD has a much wider scope, but we have focused on the most prevalent cardiac

conditions, DCM and HCM, for this analysis. The primary aim was to evaluate the diagnostic yield of testing, and the prevalence of genetic variants being identified in patients whose clinical presentations and background prompted testing but were not fully captured within the NGTD criteria.

METHOD

We analysed available results for all patients who underwent GT in the Essex ICC clinic between January 2023 and January 2025. Individuals who underwent predictive testing or clinical screening alone were excluded. All patients with DCM and HCM were categorised into two groups: fully met the criteria and did not fully meet the criteria.

Patient and public involvement

Patients/public were first involved in this research through discussion in the Essex ICC Clinic Patient and Family Support Group; approval of the study objectives was confirmed, and dissemination of the peer-reviewed findings will be via this forum.

Statistical analysis

Descriptive statistics were used to summarise patient demographics and baseline characteristics. All statistical analysis was performed using Microsoft Excel. Continuous variables were reported as means $\pm$ SD or medians with IQR, and categorical variables were expressed as counts/frequencies and percentages. Continuous data were compared using a two-sample t-test, and categorical data were compared using the χ^2 test. Gene-positive is defined as the detection of a pathogenic or likely pathogenic variant; therefore, variants of uncertain significance were excluded for the purpose of analysis.

RESULTS

A total of 257 patients with DCM and HCM had results available, totalling 136 patients with DCM (52.9%) and 121 patients with HCM (47.1%) (table 1). Out of the total cohort, the diagnostic yield from GT was 18.7%.

DCM subgroup

Of the 136 DCM patients, 84 (61.8%) were men and 52 (38.2%) were women. The median age of this cohort was 56 (IQR 43.5–64.0) years. The median age at diagnosis was 52 (IQR 39.8–60.0) years. 103 (75.7%) patients met the NGTD testing criteria, while the remaining 33 (24.3%) did not. In total, 27 patients with DCM (19.9%) were gene-positive (ie, the diagnostic yield), and the most common gene found was titin (*TTN*) (n=16, 59.3% of gene-positive results in DCM). Other genes identified included lamin A/C (*LMNA*) (n=3), myosin heavy chain 7 (*MYH7*) (n=3), actin alpha cardiac muscle 1 (*ACTC1*) (n=2), filamin-C protein (*FLNC*) (n=1), myosin-binding protein C3 (*MYBPC3*) (n=1), sodium channel type V (*SCN5A*) (n=1) and desmoglein 2 (*DSG2*) (n=1). Four

patients (14.8%) who were gene-positive did not meet current testing criteria (summarised in figure 2A).

HCM subgroup

Of the 121 HCM patients, 88 (72.7%) were men, and 33 (27.3%) were women. The median age of this cohort was 60 (IQR 52.0–69.0) years. The median age at diagnosis was 54 (IQR 46.0–65.0) years. The median age of symptom onset was 53 (IQR 45.0–63.0) years; higher in comparison to the DCM group ($p<0.05$). The mean LV wall thickness was 22 ± 4.2 mm. 74 patients with HCM (61.2%) met the current NGTD testing criteria, while the remaining 47 (38.8%) did not. In total, 21 (17.4%) patients with HCM were gene-positive, with the most common genes found being *MYBPC3* (n=14, 66.7% of the gene-positive results in HCM) and *MYH7* (n=3, 14.3%). Other genes found include troponin I3 cardiac Type (*TNNI3*) (n=2), *TTN* (n=1), alpha-protein kinase 3 (*ALPK3*) (n=1) and potassium voltage-gated channel subfamily E regulatory subunit 1 (*KCNE1*) (n=1). Three gene-positive patients with HCM (14.3%) did not meet the current testing UK NGTD criteria (summarised in figure 2B).

Screening of the seven gene-positive, yet NGTD criteria negative, DCM/HCM patients has identified 28 first-degree relatives at risk and appropriate for predictive GT (median four relatives per proband, range 2–8).

DISCUSSION

While the majority of DCM and HCM patients met the UK NGTD criteria for GT, a notable proportion (1 in 7) of the gene-positive results arose from testing outside the current iteration (14.8% in DCM versus 14.3% in HCM; $p=0.96$).

Proposal

The clinical implications of a positive genetic diagnosis cannot be understated. An abnormal genetic result does not always affect management of the patient, but it may inform risk stratification, with early consideration for intervention if required (eg, primary prevention implantable cardioverter defibrillator in *LMNA* or *FLNC* variants). A genetic diagnosis will also trigger further cascade screening, with predictive GT when appropriate and surveillance of high-risk relatives.

The ESC have reported its criteria for the testing of patients who have cardiomyopathy.⁴ Important recommendations for testing include:

- ▶ Those who fulfil phenotypical criteria of cardiomyopathy, where it benefits their clinical management or enables cascade screening.
- ▶ Those who fulfil phenotypical criteria of cardiomyopathy, where it benefits the patient struggling to understand their disease psychologically (even if it has no benefit in clinical management or for cascade screening), also taking into account patient preference.

Table 1 Patient demographics and genetic testing results

Category		Total (n=257)	HCM (n=121)	DCM (n=136)	P value
Demographics	Age (years)	57 (48.0–67.0)	60 (52.0–69.0)	56 (43.5–64.0)	<0.001
	Men	172 (66.9%)	88 (72.7%)	84 (61.8%)	0.062
Ethnicity	White	210 (81.7%)	88 (72.7%)	122 (89.7%)	<0.001
	Black/Black British	23 (8.9%)	8 (6.6%)	15 (11.0%)	0.216
	Asian/Asian British	14 (5.4%)	11 (9.1%)	3 (2.2%)	0.015
	Mixed	7 (2.7%)	5 (4.1%)	2 (1.5%)	0.259
	Unknown	3 (1.2%)	1 (0.8%)	2 (1.5%)	1.000
Comorbidities	Hypertension	108 (42.0%)	59 (48.7%)	49 (36.0%)	0.039
	Diabetes mellitus	44 (17.1%)	17 (14.0%)	27 (19.9%)	0.218
	CKD (eGFR <60)	41 (16.0%)	17 (14.0%)	24 (17.6%)	0.432
	Previous MI	10 (3.9%)	5 (4.1%)	5 (3.7%)	1.000
	BMI (kg/m ²)	29.7 (25.8–32.9)	29.2 (26.2–33.3)	29.6 (25.5–33.0)	0.609
	Current smoker	24 (9.3%)	14 (11.6%)	10 (7.4%)	0.246
	LVEF ≥45%	148 (57.6%)	100 (82.6%)	48 (35.3%)	<0.0001
	LVEF <45%	109 (42.4%)	21 (17.4%)	88 (64.7%)	<0.0001
Presenting symptoms	Shortness of breath	96 (37.4%)	22 (18.2%)	74 (54.4%)	<0.0001
	Chest pain	47 (18.3%)	24 (19.8%)	23 (16.9%)	0.545
	Palpitations	44 (17.1%)	19 (15.7%)	25 (18.4%)	0.569
	Dizziness	7 (2.7%)	4 (3.3%)	3 (2.2%)	0.710
	Syncope	18 (7.0%)	10 (8.3%)	8 (5.9%)	0.455
	OOHCA	13 (5.1%)	5 (4.1%)	8 (5.9%)	0.523
	Incidental	45 (17.5%)	33 (27.3%)	12 (8.8%)	<0.001
Genetic yield	Genotype-positive	48 (18.7%)	21 (17.4%)	27 (19.9%)	0.262
Genes identified	<i>TTN</i>	17 (6.6%)	1 (0.8%)	16 (11.8%)	
	<i>MYBPC3</i>	15 (5.8%)	14 (11.6%)	1 (0.7%)	
	<i>MYH7</i>	6 (2.3%)	3 (2.5%)	3 (2.2%)	
	<i>LMNA</i>	3 (1.2%)		3 (2.2%)	
	<i>ACTC1</i>	2 (0.8%)		2 (1.5%)	
	<i>TNNI3</i>	2 (0.8%)	2 (1.7%)		
	<i>DSG2</i>	1 (0.4%)		1 (0.7%)	
	<i>KCNE1</i>	1 (0.4%)	1 (0.8%)		
	<i>SCN5A</i>	1 (0.4%)		1 (0.7%)	
	<i>ALPK3</i>	1 (0.4%)	1 (0.8%)		
	<i>FLNC</i>	1 (0.4%)		1 (0.7%)	

Number (%) or mean (±SD) or median (IQR).

Two individuals each carried pathogenic variants in two genes; therefore, the total number of variants (n=50) exceeds the number of genotype-positive patients (n=48). P values reflect the comparison between the HCM and DCM groups.

Incidental = diagnosis made during baseline or work-up investigations (eg, murmur, electrocardiography), rather than in the context of symptom presentation.

Titin (*TTN*), myosin-binding protein C3 (*MYBPC3*), myosin heavy chain 7 (*MYH7*), lamin A/C (*LMNA*), actin alpha cardiac muscle 1 (*ACTC1*), troponin I3 cardiac type (*TNNI3*), desmoglein 2 (*DSG2*), potassium voltage-gated channel subfamily E regulatory subunit 1 (*KCNE1*), sodium channel type V (*SCN5A*), alpha-protein kinase 3 (*ALPK3*), filamin-C protein (*FLNC*).

BMI, body mass index; CKD, chronic kidney disease; DCM, dilated cardiomyopathy; eGFR, estimated glomerular filtration rate; HCM, hypertrophic cardiomyopathy; LVEF, left ventricular ejection fraction; MI, myocardial infarction; OOHCA, out of hospital cardiac arrest;

- Those with a borderline phenotype, not enough to fulfil phenotypical criteria, after detailed assessment by specialist teams.

Although the ESC sets a broader framework for testing than the UK NGTD, most of the recommendations are based on level B or C evidence (derived from a single

A: Dilated Cardiomyopathy (DCM)								
	Gene	Left Ventricular End-Diastolic Diameter (LVEDD) > 2 S.D.	Ejection Fraction < 45%	Age of Onset < 65 years	Conduction Defect	Myocardial fibrosis	Ventricular Arrhythmia	Relative with confirmed DCM/ACM
Patient 1	FLNC	✗	✗	✓	✗	✓	✗	✗
Patient 2	MYBPC3	✗	✗	✗	✗	✓	✗	✗
Patient 3	ACTC1	✗	✗	✗	✗	✓	✗	✗
Patient 4	TTN	✗	✗	✗	✗	✓	✗	✗
B: Hypertrophic Cardiomyopathy (HCM)								
	Gene	Wall Thickness ≥ 15 mm	Age of Onset < 60 years	Relative with confirmed HCM	Wall Thickness ≥ 13 mm			
Patient 1	MYBPC3	✓	✗	✗	✓			
Patient 2	ALPK3	✗	✓	✗	✓			
Patient 3	MYH7	✓	✗	✗	✓			
✓ = Criteria met ✗ = Criteria not met								

Figure 2 Overview of the gene-positive patients who did not meet the UK NGTD criteria for genetic testing. (A) DCM patients with pathogenic variants not fulfilling NGTD criteria. (B) HCM patients with pathogenic variants not fulfilling NGTD criteria. Filamin-C protein (*FLNC*), myosin-binding protein C3 (*MYBPC3*), actin alpha cardiac muscle 1 (*ACTC1*), titin (*TTN*), alpha-protein kinase 3 (*ALPK3*), myosin heavy chain 7 (*MYH7*). NGTD, National Genomic Test Directory.

randomised/large non-randomised study or formed from a consensus of expert opinion, respectively); this highlights the paucity of data to guide testing algorithms. However, we hope the data reported in this study will prompt further discussion and review, with adaptation of the guidelines and directories encouraged.

Our opinion is that an age cut-off in the absence of family history is punitive, given that 3 out of the 4 (75.0%) gene-positive but NGTD-criteria-negative patients with DCM had an age of onset above 65 years (median age was 56 (IQR 43.5–64.0)). A similar pattern was seen in the HCM cohort, with 2 out of 3 gene-positive but NGTD-criteria-negative patients (66.6%) presenting with an age of onset above 60 years, strengthening our argument against the current age-specific criteria.

While the NGTD remains a valuable tool for national standardisation, its future use may benefit from incorporating broader clinical discretion focusing on phenotypic-driven criteria. There are potential cost implications to wider testing,^{7,8} but our data support a role for redefined testing criteria and consideration of bespoke subpanels with higher yield and potentially better cost-effectiveness. The newly defined NDLCV with proven myocardial scarring warrants consideration of a dedicated yet smaller subpanel including *LMNA*, *FLNC*, phospholamban (*PLN*) and desmin (*DES*) gene variants, potentially without the need to include all variants associated with LV dilatation; further discussion between clinicians and genomic laboratories is encouraged to define the variants to be included. Similarly, consideration of a myocarditis subpanel including desmoplakin (*DSP*) and *DSG2* gene variants is recommended, given the potential

for ACM to mimic myocarditis in the ‘hot phase’.⁹ There is a tendency to attribute myocarditis to viral infections; however, the recent 2025 ESC guidelines recommend GT in patients with recurrent myocarditis, phenotypically mild cardiomyopathies in the context of a family history of myocarditis, or myocarditis presenting with severe ventricular arrhythmias.¹⁰

The authors acknowledge that the burden on the UK genomics laboratories is not purely financial, that is, the cost of testing. An increase in demand has resulted in the development of a significant backlog, affecting the turnaround time for processing and results reporting. Any further increase in demand due to wider testing criteria will affect the timeliness of reporting, and this could have clinical implications. There is a finite capacity in the UK NGTD-affiliated laboratories, however we hope the suggested amendments will streamline pathways, allowing more targeted testing and achievement of the 42-day target. We also accept that the genomics laboratories are always willing to discuss individual cases and proceed with testing in cases outside the criteria that have been discussed in a wider multidisciplinary team setting; however, requests sent routinely, without discussion, may be rejected based on the UK NGTD alone.

Gene elusive patients may benefit from future wider GT, either via retesting with panels including novel variants or as genome-wide association studies. However, genotyping all patients attending ICC clinics without some inclusion and exclusion criteria is clearly inappropriate, given the cost implications, effect on turnaround time of results reporting, and significant risk of identifying variants of uncertain significance with no clinical

relevance and causing unnecessary distress to patients and their relatives.

There is a process of predictive testing in families with pathogenic or likely pathogenic gene variants to detect relatives at risk. The absolute number of gene-positive patients not meeting the current NGTD criteria in this study was low; however, this resulted in the identification of 28 individuals, an average of 4 first-degree relatives per proband, appropriate for predictive GT. While a positive predictive genetic test enables early introduction of prognostic therapies to reduce the risk of future adverse events, it is equally beneficial to the patient and cost-saving for the NHS when first-degree relatives are subsequently found to be gene negative, removing the need for life-long clinical screening.

Limitations

This is a study of patients who were referred for GT mainly based on clinical suspicion of inherited disease; we acknowledge that the retrospective analyses and reinterpretation of testing criteria are likely to include significant selection bias. However, within these limitations, the need to re-evaluate the current testing criteria is justified by the results reported. We have not reported data regarding patients deemed inappropriate for genotyping, and we accept this group may also include gene-positive patients with implications for wider screening.

The yield of GT overall was lower than might be expected in a tertiary referral centre. The potential reason for the low yield is referral from a relatively new ICC service, established for only 2 years at the time of data collection, which is continually defining and evolving its local criteria for requesting genetic tests. The analysis of the genotyped patients presented in this work is likely to improve the gene-positive rates.

CONCLUSION

This analysis underscores the important role of GT in the diagnosis and management of cardiomyopathies. While the NGTD provides a structured and detailed framework to triage patients needing further testing, our data suggest the current criteria are restrictive, with risk of missing genotype-positive patients. Rigid adherence to UK NGTD criteria may prevent early diagnosis in relatives at risk, thus limiting the potential for prompt initiation of prognostic therapies or promoting unnecessary long-term clinical screening in individuals who would have benefited from a negative predictive test.

Given the increasing accessibility and affordability of GT, we believe re-evaluation of the UK NGTD criteria is indicated and should allow for greater clinical discretion or a broader criteria description. Future studies involving larger cohorts and analyses are warranted to guide areas of refinement in the testing framework. The role of smaller subpanels in specific subtypes of cardiomyopathy, including NDLVC and some myocarditis cases, is evidence-based and likely to be beneficial and potentially more cost-effective.

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Contributors ACM and JND were responsible for study conception. AHW was responsible for data provision. AG assisted with manuscript review. ACM carried out data collection, review of patient records, statistical analysis and data interpretation. ACM drafted the manuscript and conducted the research related to the discussion. ADD and HOS provided senior clinical oversight, reviewed and edited the manuscript. JND provided overall supervision, helped interpret findings and was primarily responsible for critical revision of the manuscript. All authors approved the final version of the manuscript. JND is the guarantor.

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Patient consent for publication Not applicable.

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